United States Patent Application Publication	20250275643
Kind Code	A1
Publication Date	September 04, 2025
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COFFEE BREWING APPARATUS AND METHOD

Abstract

A coffee brewing apparatus includes a housing, a grinding system associated with the housing, the grinding system having a grinding mechanism for grinding coffee beans to transform the coffee beans into coffee grounds, and a brewing system associated with the housing, the brewing system including a brewing chamber for receiving a cartridge, and a flow pathway for delivering heated water to the cartridge.

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Appl. No.: **18/591155**

Filed: **February 29, 2024**

Publication Classification

Int. Cl.: **A47J31/42** (20060101); **A47J31/06** (20060101)

U.S. Cl.:

CPC **A47J31/42** (20130101); **A47J31/0689** (20130101);

Background/Summary

FIELD OF THE INVENTION
[0001] The present invention relates generally to beverage making devices and, more particularly, to a coffee brewing apparatus and method.

BACKGROUND OF THE INVENTION

[0002] Various types of beverage brewers, such as single-serve beverage brewers, have been developed and are popular for home brewing of coffee and other beverages. In a typical single-serve beverage brewer, a disposable beverage filter cartridge containing a dry beverage medium (such as tea or coffee) is disposed within a filter cartridge holder or brewing holster of a beverage or coffee machine. To brew the beverage, the filter cartridge is pierced by inlet and outlet probes to accommodate a through-flow of metered hot water. The hot water infuses the dry beverage medium contained in the cartridge to thereby produce a single serving of the beverage. After the beverage is brewed, the cartridge is removed from the cartridge holder and it is discarded.

[0003] Existing coffee machines and single-serve cartridges, including those of the reusable variety, are typically only capable of brewing a single type of coffee such as, for example, espresso or drip coffee, but not both. Moreover, most existing coffee machines are only configured to receive a specific type of cartridge or pod, and cannot accommodate cartridges/pods from different manufacturers.

[0004] In view of the above, there is a need for a coffee brewing apparatus that is configured to accommodate a variety of single use and reusable beverage cartridges/pods of different style and/or configuration, and which has both grinding and brewing functions.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide a coffee brewing apparatus.

[0006] It is another object of the present invention to provide a coffee brewing apparatus having both grinding and brewing functions.

[0007] It is another object of the present invention to provide a coffee brewing apparatus which can accommodate a variety of single use and reusable beverage cartridges/pods of different style and/or configuration.

[0008] It is another object of the present invention to provide a coffee brewing apparatus which is configured for both high-pressure and low-pressure brewing operations.

[0009] It is another object of the present invention to provide a coffee brewing apparatus which is configured to produce a variety of coffee beverages of different types.

[0010] These and other objects are achieved by the present invention.

[0011] According to an embodiment of the present invention, a coffee brewing apparatus includes a housing, a grinding system associated with the housing, the grinding system having a grinding mechanism for grinding coffee beans to transform the coffee beans into coffee grounds, and a brewing system associated with the housing, the brewing system including a brewing chamber for receiving a cartridge, and a flow pathway for delivering heated water to the cartridge.

[0012] According to yet another embodiment of the present invention, a method for producing coffee includes the steps of positioning a cartridge beneath a grinding mechanism of a coffee brewing apparatus, actuating the grinding mechanism to transform coffee beans into coffee grounds, collecting the coffee grounds in the cartridge, moving the cartridge with coffee grounds from a position beneath the grinding mechanism to a brewing chamber within a brew head located adjacent to the grinding mechanism, and initiating a brewing operation to cause heated water to pass through the coffee grounds within the cartridge.

[0013] According to yet another embodiment of the present invention, a coffee brewing apparatus includes a housing, a grinding system located on a front the housing, the grinding system having a grinding mechanism for grinding coffee beans to transform the coffee beans into coffee grounds, a cartridge holder located beneath the grinding mechanism for holding the cartridge beneath the grinding mechanism, for collection of the coffee grounds discharged by the grinding mechanism in the cartridge, and a brew head located on the front of the housing adjacent to the grinding system, the brew head including a brewing chamber for receiving the cartridge containing the coffee grounds, the brew head being movable between an open position allowing access to the brewing chamber for insertion and/or removal of the cartridge, and a closed position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The present invention will be better understood from reading the following description of non-limiting embodiments, with reference to the attached drawings, wherein below:

[0015] FIG. **1** is a perspective, front view of a coffee brewing apparatus according to an embodiment of the present invention.

[0016] FIG. **2** is a perspective, bottom view of coffee brewing apparatus of FIG. **1**.

[0017] FIG. **3** is a front elevational view of the coffee brewing apparatus of FIG. **1**.

[0018] FIG. **4** is a left side elevational view of the coffee brewing apparatus of FIG. **1**.

[0019] FIG. **5** is a right side elevational view of the coffee brewing apparatus of FIG. **1**.

[0020] FIG. **6** is a top plan view of the coffee brewing apparatus of FIG. **1**.

[0021] FIG. **7** is a side, cross-sectional view of the coffee brewing apparatus of FIG. **1**.

[0022] FIG. **8** is a schematic illustration of a brewing system of the coffee brewing apparatus of FIG. **1**.

[0023] FIG. **9** is a cross-sectional view of a portion of the coffee brewing apparatus, showing a brewing operation.

[0024] FIG. **10** is a front elevational view of a grinding system of the coffee brewing apparatus of FIG. **1**, according to an embodiment of the invention.

[0025] FIG. **11** is a cross-sectional view of the grinding system of the coffee brewing apparatus of FIG. **1**.

[0026] FIG. **12** is a schematic illustration of the coffee brewing apparatus of FIG. **1**, showing an alternative grinding system.

[0027] FIG. **13** is a schematic illustration of the coffee brewing apparatus of FIG. **1**, showing an alternative grinding system.

[0028] FIG. **14** is a schematic illustration of the coffee brewing apparatus of FIG. **1**, showing an alternative grinding system.

[0029] FIG. **15** is a perspective, top view of a reusable coffee and beverage pod for use with the coffee brewing apparatus of FIG. **1**.

[0030] FIG. **16** is a side, cross-sectional view of the reusable coffee and beverage pod of FIG. **15**.

[0031] FIG. **17** is a cross-sectional view of a portion of the coffee brewing apparatus, illustrating use with a dual-pressure brewing pod.

[0032] FIG. **18** is a side-cross-sectional view of a portion of the coffee brewing apparatus, illustrating a locking mechanism thereof.

[0033] FIG. **19** is a top, cross-sectional view of a portion of the coffee brewing apparatus, illustrating the locking mechanism thereof.

[0034] FIG. **20** is a front elevational view of the coffee brewing apparatus of FIG. **1**, showing a dual pressure brewing pod in a grinding position.

[0035] FIG. **21** is a front elevational view of the coffee brewing apparatus of FIG. **1**, showing a dual pressure brewing pod in a brewing position.

[0036] FIG. **22** is a front elevational view of a portion of the coffee brewing apparatus, showing the grinding system in a fine grind position.

[0037] FIG. **23** is a side elevational view of the coffee brewing apparatus, illustrating a grinding operation.

[0038] FIG. **24** is a perspective view of the coffee brewing apparatus, illustrating the reusable coffee pod in the brewing position.

[0039] FIG. **25** is a perspective view of the coffee brewing apparatus, illustrating a brewing operation.

[0040] FIG. **26** is a perspective view of a K-cup pod holder for use with the coffee brewing

apparatus of FIG. **1**.
[0041] FIG. **27** is a perspective view of a Dolce Gusto pod holder for use with the coffee brewing apparatus of FIG. **1**.
[0042] FIG. **28** is a perspective view of a Nespresso pod holder for use with the coffee brewing apparatus of FIG. **1**.
[0043] FIG. **29** is a perspective view of a portion of the coffee brewing apparatus, illustrating use of different pod holders.
[0044] FIG. **30** is a perspective view of a brew head of the coffee brewing apparatus of FIG. **1**.
[0045] FIG. **31** is a side, cross-sectional view of the brew head of the coffee brewing apparatus of FIG. **1**.
[0046] FIG. **32** is another perspective view of the K-cup pod holder of FIG. **26**.
[0047] FIG. **33** is a cross-sectional view of the K-cup pod holder of FIG. **26**.
[0048] FIG. **34** is a cross-sectional view of a portion of the coffee brewing apparatus, showing use of the K-cup pod holder of FIG. **26**.
[0049] FIG. **35** is another perspective view of the Dolce Gusto pod holder of FIG. **27**.
[0050] FIG. **36** is a cross-sectional view of the Dolce Gusto pod holder of FIG. **27**.
[0051] FIG. **37** is a cross-sectional view of a portion of the coffee brewing apparatus, showing use of the Dolce Gusto pod holder of FIG. **27**.
[0052] FIG. **38** is another perspective view of the Nespresso pod holder of FIG. **28**.
[0053] FIG. **39** is a perspective view of a portion of the coffee brewing apparatus, showing use of the Nespresso pod holder of FIG. **28**.
[0054] FIG. **40** is a cross-sectional view of a portion of the coffee brewing apparatus, showing use of the Nespresso pod holder of FIG. **28**.
[0055] FIG. **41** is another cross-sectional view of a portion of the coffee brewing apparatus, showing use of the Nespresso pod holder of FIG. **28**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0056] With reference to FIGS. **1-7**, a coffee brewing apparatus **10** according to an embodiment of the present invention is shown. The coffee brewing apparatus includes a generally rectangular housing **12**, a grinding system having a grinding head **14** supported by the housing **12**, and a brew head **16** of a brewing system supported by the housing **12**. As shown in FIGS. **1** and **3**, for example, in an embodiment, the grinding head **14** and brew head **16** are arranged adjacent to one another on a front of the housing **12**. A drip tray **18** having a slotted upper surface extends forwardly from the housing **12** at a lower portion of the housing **12**, and is configured to receive a cup or mug thereon for receiving a brewed beverage from the brew head **16**. The housing **12** includes an access door or panel **20** on one side thereof, permitting access to internal components of the apparatus **10**, and includes a reservoir **22** for holding a volume of water for use in a brewing operation, as described in detail below.
[0057] As best shown in FIGS. **3**, **8** and **9**, the primary components of the grinding system/grinding head **14** are shown, and include a hopper **24** having a removable lid **25**, a motor **27**, and a grinding mechanism such as, for example, a burr grinder **26** mounted beneath the motor **27** and operatively connected to the motor **27** so as to be rotatably driven by the motor **27**, and a cartridge holder **28** extending forwardly from the housing **12** beneath the burr grinder **26**. In an embodiment, the cartridge holder **28** includes a pair of arms **30** that extend forwardly from the housing **12**, which are configured to receive a reusable pod or cartridge (not shown).
[0058] As best shown in FIG. **7**, the primary components of the brewing system **32** of the coffee brewing apparatus are shown, and include the water reservoir **22** for containing a volume of water, a water pump **34**, a heater **36** and an air pump **38**. As shown in FIG. **8**, the brewing system **32** further includes the brew head **16** and brewing chamber **40** (which receives a reusable or disposable pod or cartridge containing one or more dry beverage ingredients). In particular, the brew head **16** is moveable between an open position where the brewing chamber **40** is accessible to

remove or insert a beverage pod or cartridge, and a closed position, as discussed in detail hereinafter).As also shown therein, the air pump is fluidly connected to the brew head **16** via a flow line **42** having a check valve **44**. The water reservoir **22** is likewise fluidly connected to the brew head **16** via flow line **46** that extends between the reservoir **22** and heater **36**, and flow line **48** that extends from the heater **36** to the brew head. Check valve **50** is positioned along flow line **48** for controlling a flow of heated water therethrough. In operation, water is pumped from the reservoir **22** by pump **34** to the heater **36** where it is heated to a target temperature. The heater **36** may be any type of heater known in the art, such as an in-line heater. Hot water then flows through flow line **48**, across check valve **50**, and to the brew head **16**. The water pump **34** pushes the hot water through coffee grounds or other beverage ingredients within the brewing chamber **40**, producing brewed coffee which exits through an opening **52** in the bottom of the brewing chamber **40**, as illustrated in FIG. **9**. In an embodiment, the air pump **38** is then actuated to pump air into the brewing chamber **40** to dry out the brewing chamber **40**.

[0059] Turning now to FIGS. **10** and **11**, the configuration and operation of the grinding system/grinding head **14** is shown. As shown therein, and as disclosed above, the grinding head **14** includes a hopper **24** for receiving for example, whole coffee beans **54**, a motor **27**, and a grinding mechanism in the form of a burr grinder **26** rotatably driven by the motor **27**. In operation, whole coffee beans **54** or other ingredients to be processed are loaded into the hopper **24**. The beans **54** follow the slide into the burr grinder **26**. The motor **27** drives the burr grinder **26** to grind the coffee beans **54** into coffee grounds **60** which exit the grinding head **14** via an outlet **56** on an underside of the burr grinder **26**. The coffee grounds are then collected by a reusable beverage pod or cartridge **58** received by the cartridge holder **28** beneath the burr grinder **26**. In an embodiment, the burr grinder **26** is connected to the grinder motor **27** by a clutch structure, which allows the assembly to be removed and/or disassembled for cleaning. In an embodiment, the grinding head **14** also includes an adjustment ring **29** which allows a user to adjust the coarseness or fineness of the coffee grounds (by adjusting the spacing between the burrs of the burr grinder **26**).

[0060] While FIGS. **10** and **11** illustrate the grinding head **14** as including a conical burr grinder **26**, the present invention is not intended to be so limited in this regard. Rather, the grinding head **14** may have one of a variety of different types of grinding mechanisms. In particular, with reference to FIGS. **12**, **13** and **14**, the grinding head of the coffee brewing apparatus may include a conical burr grinder **26**, as described above in connection with FIGS. **10** and **11**, and as illustrated schematically in FIG. **12**. Alternatively, the grinding head **14** may include a flat burr grinder **62**, which has a horizontally-oriented motor **64**, and a hopper **66** that funnels ingredients to the flat burrs of the flat burr grinder **62**. Moreover, the grinding head **14** may instead include a blade-style grinder **68** having a hopper **70**, a blade **72** rotatably mounted within the hopper **70**, and a motor **74** beneath the hopper **70** for rotatably driving the blade **72** to grind coffee beans within the hopper.

[0061] In an embodiment, the brew head **16** is configured to accommodate a variety of different single-serve pods or cartridges containing beverage ingredients such as ground coffee. For example, in one embodiment, the brew head **16** is configured to receive a reusable, single-serve, dual pressure beverage pod of the type shown and disclosed in U.S. patent application Ser. No. 18/591,085, filed on Feb. 29, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety. FIGS. **15** and **16** illustrate such a reusable, single-serve, dual pressure beverage pod **58**, according to one embodiment of the invention. As shown therein, the beverage pod includes a generally cup shaped body **92** sized and configured to be received in a filter cartridge holder of a beverage or coffee machine (such as coffee brewing apparatus **10**). The body **92** includes a generally cylindrical upper chamber **94**, a generally conical lower chamber **96**, and an actuator in the form of a rotatable control ring **98** intermediate, and interconnecting, the upper chamber **94** and the lower chamber **96**. The reusable coffee and beverage pod **58** includes a lid **100** pivotally connected to the upper chamber **94** about hinge **102**. The lid **100** encloses the open end of the upper chamber **94** and includes an inlet opening or aperture **104** coaxial with the upper and lower chambers **94**, **96** for the ingress of heated water. The lower chamber **96** includes an outlet opening or aperture **106** at the lower end thereof and coaxial with the upper and lower chambers **94**, **96**. A tab **108** extends from one side of the body **92** adjacent to the hinge **102** and is graspable by a user to manipulate the pod **58**, such during removal and insertion of the pod **58** with respect to the beverage or coffee machine **19**.

[0062] As illustrated in FIG. **16**, the internal configuration of the coffee and beverage pod **58** is shown. In an embodiment, the coffee and beverage pod **58** includes a sealing element **110** such as an O-ring or the like that forms a substantially water-tight seal between the lid **100** and the upper rim of the upper chamber **94** of the body **92** when the lid **100** is in the closed position. The coffee and beverage pod **58** also includes a shower head **112** that is configured distribute water evenly, via an array of outlets, through coffee grounds or other beverage ingredient placed within the upper chamber **94**, as disclosed below. In an embodiment, the sealing element **110** and shower head **112** are integrated with the lid **100**. As also shown therein, the upper chamber **94** includes a generally conical bottom having an outlet **114** in fluid communication with the lower chamber **96**, which similarly has a generally conical bottom having outlet **106**. In an embodiment, a lower portion of the outlet **106** has the shape of an inverted cone (having a diameter at a lower end that is greater than a diameter at a location above the lower end). A generally cylindrical filter element **116** is located within the upper chamber **94** and is configured to retain a beverage ingredient such as coffee grounds, tea leaves or the like, but allows for the passage of liquid therethrough. In an embodiment, the filter element **94** may take a variety of forms such as, for example, a fine mesh screen, a porous fabric, or the like. As disclosed more fully in U.S. patent application Ser. No. 18/591,085, the beverage pod **58** also includes an adjustable pressure valve **118** that is moveable between open and closed position (closed position shown in FIG. **16**) by actuation of the control ring **98**. In particular, in the closed position, the valve **118** allows high pressure (5-12 bar) to build up in the upper chamber **94** before pushing the valve **118** downward to release brewed coffee, while in the open position, the valve does not obstruct outlet **114** and thus coffee is permitted to flow freely from upper chamber **94** to lower chamber **96**.

[0063] FIG. **17** shows the reusable, dual pressure beverage pod **58** received in the brewing chamber **52** of the brew head **16** of the coffee brewing apparatus **10**. As shown therein, after inserting the beverage pod **58**, the brew head **16** is closed, and the beverage pod **58** is locked tightly inside the brewing chamber **52** of the brew head **16**. In this position, an annular sealing element **76** in the brew head **16** is compressed and sealed tightly against the lid **100** of the beverage pod **58**. In this position, a water outlet **80** in the brew head **16** is in fluid communication with a shower head inlet **104** in the lid **100** of the beverage pod **58** for supplying water thereto (which provides for the flow of water to the shower head **112** of the beverage pod **58**). Importantly, the sealing element **76** ensures that heated water from the water outlet **80** of the brew head **16** passes through the upper chamber **94** of the beverage pod **58** during brewing.

[0064] With reference to FIGS. **18** and **19**, the brew head **16** includes a locking mechanism that is designed to withstand high pressures generated in the beverage pod **58** during the brewing operations (e.g., espresso brewing), but which also allows for easy beverage pod/cartridge exchange. As shown therein, the locking mechanism includes a locking lever **120** accessible from atop the brew head **16**, which is operatively connected inside the upper portion **122** of the brew head **16** to a cam and follower structure. In particular, the lever **120** may be generally U-shaped and is operatively connected at respective ends thereof to a pair of followers **126** via respective cam members **128**. The locking mechanism further includes opposed hooks or retaining members **130** mounted to the lower portion **124** of the brew head **16**. In operation, after inserting a beverage cartridge/pod (e.g., reusable pod **58**) in the brewing chamber **52** of the lower portion **124** of the brew head **16**, the locking lever **120** is pushed downwardly, causing the followers **126** to move downwardly and slide beneath the respective retaining members **130**, as best shown in FIG. **18**. In this position, the retaining members **130** cinch or retain the followers **126** to maintain the upper portion **122** of the brew chamber **16** in contact with the lower portion **124** (and maintain the sealing element **110** in sealing contact with the lid of the beverage pod **58**. As shown in FIG. **18**, the retaining members **130** and followers **126** align on an axis **132** passing through the center of brewing chamber **52**, which provides the greatest counter force against the brewing pressure (i.e., the pressure that builds up within the beverage pod **58**). After a brewing operation, the lever **120** can be moved upwardly, which moves the followers **126** out if engagement with the retaining members **130**, enabling the upper portion **122** of the brew head **16** to pivot to an open position, allowing for the beverage pod **58** to be removed from the brewing chamber **40**.

[0065] With reference to FIGS. **20** and **21**, the coffee brewing apparatus **10** of the present invention combines two independently grinding and brewing systems to provide an integrated appliance. As shown in FIG. **20**, a reusable beverage pod, such as beverage pod **58**, can first be placed on cartridge holder **28** beneath the grinding head **14**. Whole coffee beans are loaded into hopper **24**, and processed into coffee grinds using burr grinder **26**. The coffee grounds exit through outlet **56** and are collected in the upper chamber **94** of the beverage pod **58**. After receiving an adequate amount of coffee grounds, the grinder stops automatically. The beverage pod **58** containing coffee grounds is then moved to the brew head **16** and placed in the brewing chamber in the lower portion **124** of the brew head **16**, as shown in FIG. **21**. A brewing operation can then be carried out to produce a brewed beverage (e.g., drip-style coffee or espresso), which is then collected in a cup or mug **134** placed beneath the brew head **16**. Importantly, the grinding and brewing systems run independently, but are integrated into a single machine.

[0066] Turning now to FIGS. **22-25**, the grinding system **14** may have a plurality of grinding settings corresponding to different grind sizes including, for example, a first grind setting corresponding to a coarser grind size for drip-style coffee, and a second grind setting corresponding to a finer grind size for espresso. As illustrated in FIG. **22**, for example, adjustment ring **29** can be rotated to the espresso setting to produce a finer grind size suitable for espresso. At the same time, the control ring **98** of the reusable beverage pod **58** can be rotated to the high-pressure (e.g., espresso) brewing mode, causing the adjustable valve **118** to move to the closed position. As discussed above, the grinding system **14** is then actuated to controllably dispense an appropriate amount of fine coffee grounds into the beverage pod **58**, as shown in FIG. **23**. The beverage pod **58** is then moved to the brew head **16**, as discussed above, and as shown in FIG. **24**. The brewing head **16** is then closed, and an espresso brewing mode is then actuated (e.g., by a user depressing a button) to initiate brewing. The coffee brewing apparatus **10** then brews a cup of espresso according to a program stored in memory, as shown in FIG. **25**. During such mode, the high-pressure pump **34** may be run at full speed, which produces a high brewing pressure (approximately 5-12 bar) within the beverage pod **58** for a short time for producing espresso. As illustrated in FIG. **25**, in an embodiment, at least a portion of the drip tray **18** may be slidably mounted on the housing **12** so that a location of the drip tray **18** relative to the outlet of the brew head **16** can be adjusted, so as to prevent splattering when using smaller cups.

[0067] Alternatively, for producing a drip-style cup of coffee, the reusable beverage pod **58** is placed beneath the grinding head **14** and the adjustment ring **29** on the grinding head **14** is be rotated to the drip setting to produce a coarser grind size suitable for drip style coffee. At the same time, the control ring **98** of the reusable beverage pod **58** can be rotated to the low-pressure (e.g., drip coffee) brewing mode, causing the adjustable valve **118** to move to the open position. As discussed above, the grinding system **14** is then actuated to controllably dispense an appropriate amount of coarse coffee grounds into the beverage pod **58**. The beverage pod **58** is then moved to the brew head **16**, as discussed above. The brewing head **16** is then closed, and a drip coffee brewing mode is then actuated (e.g., by a user depressing a button) to initiate brewing. The coffee brewing apparatus **10** then brews a cup of drip style coffee according to a program stored in memory. During such mode, the high-pressure pump **34** may be run at a reduced speed, which produces a lower brewing pressure (less than approximately 2 bar) within the beverage pod **58** for a

longer time for producing drip-style coffee.
[0068] Turning now to FIGS. **26-41**, the coffee brewing apparatus **10** is also configured to receive a variety of different single-serve coffee pods from different manufacturers, in addition to reusable pod **58**. For example, as shown in FIGS. **26**, **27** and **28**, the coffee brewing apparatus **10** includes at least three interchangeable capsule holders **200**, **300**, **400** that are configured to receive K-cups, Dolce Gusto pods, and Nespresso capsules, respectively. As shown therein, the capsule holders **200**, **300**, **400** include a cup shaped body **202**, **302**, **402**, a generally open top **204**, **304**, **404**, an outlet **206**, **306**, **406** at a bottom thereof, and a grasping tab **208**, **308**, **408**. In each case, the cup shaped body **202**, **302**, **402** is sized and dimensioned to receive a K-cup, Dolce Gusto pod, or Nespresso capsule, respectively. Moreover, the capsule holders **200**, **300**, **400** are sized and dimensioned to be received in the brewing chamber **40** in the brew head **16** in a manner similar to reusable beverage pod **58**, as shown in FIG. **29**.
[0069] In use, changing the capsule holders involves, opening the brew head **16** by pulling on the locking lever **120** to unlock the system and open the brew head cover, exchange the holder by placing one of the capsule holders **200**, **300**, **400** or the dual pressure brewing pod **58** into the brewing chamber **40**, and closing and locking the brew head **16** for brewing. This simple drop-in design provides easy switching between different capsule designs and allows for various pressure brewing.
[0070] With reference to FIGS. **30** and **31**, the underside of the upper portion **122** of the brew head **16** is shown, and includes a piercing mechanism **138** (which may be, for example, a 3-blade array) which are configured to cut open the capsule received in the capsule holder when the upper portion **122** of the brew head **16** is moved to the closed position and the locking lever **120** moved to the locked position. In an embodiment, all three types of capsules share a common piercing method. Importantly, as the upper portion **122** of the brew head **16** is moved to the locked position, the sealing element **76** contacts the capsule or capsule holder **200**, **300**, **400** to create a fluid tight seal, preventing the leak of how water from the rim of capsule holder, and ensure the hot water is pushed into the capsule for brewing. As also shown therein, the underside of the upper portion **122** of the brew head **16** also includes the water outlet **80**, which provides hot water to the capsule after piercing.
[0071] FIGS. **32-34** more clearly illustrate the configuration of the K-cup capsule holder **200** and the manner in which it receives a K-cup capsule **220** and interfaces with the coffee brewing apparatus **10**. As best shown in FIG. **32** the interior of the capsule holder **200** includes a hollow needle **210** for piercing the bottom of the K-cup capsule **220** when it is received therein. The needle **210** includes an annular sealing element **212** that is configured to create a seal between the needle **210** and the bottom of the K-cup capsule **220** when the needle **210** penetrates the capsule **220**. Beneath the needle **210** is a lower chamber **214** in fluid communication with the interior passage of the needle **210** for the collection of coffee. As shown in FIG. **34**, after inserting a K-cup capsule **220** within the capsule holder **200**, the capsule holder **200** is inserted into the brewing chamber **40** within the lower portion **124** of the brew head **16** and the brew head **16** is moved to the locked position. The piercing blade **138** penetrates the capsule foil **222** to create holes for water to flow in for brewing, and the needle **210** at the bottom of the capsule holder **200** penetrates the bottom part of the capsule to create the outlet for the beverage to flow out. Importantly, the sealing element **76** is sealed with the capsule foil **222**. During brewing, the water from the water outlet **80** is forced to pass through the hole created by the piercing blade **138** in the foil **222**, where it passes through coffee grounds **224** within a filter basket **226** within the capsule **220**, creating brewed coffee which exits through needle **210**, collects in bottom chamber **214**, and exits through outlet **206**.
[0072] FIGS. **35-37** more clearly illustrate the configuration of the Dolce Gusto capsule holder **300** and the manner in which it receives a Dolce Gusto capsule **320** and interfaces with the coffee brewing apparatus **10**. As shown in FIG. **37**, after inserting a Dolce Gusto capsule **320** within the capsule holder **300**, the capsule holder **300** is inserted into the brewing chamber **40** within the

lower portion **124** of the brew head **16** and the brew head **16** is moved to the locked position. The piercing blade **138** penetrates the capsule foil **322** to create holes for water to flow in for brewing. Importantly, the sealing element **76** is sealed with the capsule foil **322**. During brewing, the water from the water outlet **80** is forced to pass through the hole created by the piercing blade **138** in the foil **322**, where it passes through coffee grounds **324** within the capsule **320**. During this process, the pressure generated inside the capsule **320** presses down a foil layer **310** within the capsule **300**, causing it to contact a punching bed **312** within the capsule **300**, creating an outlet in the bottom of the capsule **300** through which the brewed coffee can exit the capsule **300**. The brewed coffee then exits through outlet **306** in the capsule holder **300**.

[0073] FIGS. **38-41** more clearly illustrate the configuration of the Nespresso capsule holder **400** and the manner in which it receives a Nespresso capsule **420** and interfaces with the coffee brewing apparatus **10**. As best shown in FIGS. **38** and **39**, the capsule holder **400** includes a cover **410** pivotally connected to the body **402**, and which includes a chamber **412** having an annular sealing element **414**. The floor of the body **402** includes a piercing bed **416** having a plurality of piercing elements, and allows for the flow of fluid therethrough. The cover **410** is moveable between open and closed positions. As shown in FIG. **39**, in the open position, a Nespresso capsule **420** can be received in the chamber **412**. As shown in FIG. **41**, after inserting a Nespresso capsule **420** within the capsule holder **400**, the capsule holder **400** is inserted into the brewing chamber **40** within the lower portion **124** of the brew head **16** and the brew head **16** is moved to the locked position. The piercing blade **138** penetrates the bottom of the capsule **420** (which is in the inverted position) to create holes for water to flow in for brewing. Importantly, the sealing element **76** is sealed with the capsule **420**. During brewing, the water from the water outlet **80** is forced to pass through the hole created by the piercing blade **138** in the capsule **420**, where it passes through coffee grounds **424** within the capsule **420**. During this process, the pressure generated inside the capsule **420** presses down the foil cover **426** of the capsule, causing it to contact the piercing/punching bed **416** of the capsule holder **400**, creating an outlet in the bottom of the capsule **400** through which the brewed coffee can exit the capsule **400**. The brewed coffee then exits through outlet **406** in the capsule holder **400**.

[0074] The coffee brewing apparatus **10** of the present invention therefore provides an easy to use system that allows for brewing of coffee beverages from whole coffee beans, or from a variety of different single-serve capsules/cartridges. The use of the reusable beverage pod also allows for the brewing of different style beverages (e.g., drip style coffee or espresso), depending on user preference. In connection with the above, the coffee brewing apparatus **10** includes a user interface, which may include a display and a variety of knobs, buttons or the like that enable a user to provide power to the appliance, select grinding and brewing functions, initiate grinding, initiate brewing, and turn off the appliance. The coffee brewing apparatus **10** additionally includes a controller configured to carry out a variety of preprogrammed grinding and brewing modes in dependence upon user selection, and is configured to control operation of, at least, the air and water pumps, heater, and grinder. Importantly, the side-by-side configuration of the grinding head **14** and brew head **16** allows a user to carry out grinding and brewing operations simultaneously, using a single reusable beverage pod **58**. However, the coffee brewing apparatus **10** also allows for grinding and brewing operations to be carried out separately and independently. Accordingly, the coffee brewing apparatus **10** of the present invention provides for an ease of use and level of functionality and interchangeability heretofore not seen in the art.

[0075] Although this invention has been shown and described with respect to the detailed embodiments thereof, it will be understood by those of skill in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiments disclosed in the above detailed description, but that the invention will include all embodiments falling within the scope of this disclosure.

Claims

1. A coffee brewing apparatus, comprising: a housing; a grinding system associated with the housing, the grinding system having a grinding mechanism for grinding coffee beans to transform the coffee beans into coffee grounds; and a brewing system associated with the housing, the brewing system including a brewing chamber for receiving a cartridge, and a flow pathway for delivering heated water to the cartridge.
2. The coffee brewing apparatus of claim 1, wherein: the grinding system and the brewing system are located on a front side of the housing and are positioned adjacent to one another.
3. The coffee brewing apparatus of claim 2, wherein: the grinding system includes an outlet for discharging the coffee grounds; and wherein the coffee brewing apparatus further includes a cartridge holder beneath the outlet of the grinding system for holding the cartridge beneath the outlet, for collection of the coffee grounds discharged by the outlet in the cartridge.
4. The coffee brewing apparatus of claim 3, wherein: the cartridge is a reusable beverage pod.
5. The coffee brewing apparatus of claim 4, wherein: the reusable beverage pod is selectively configurable in both a high-pressure brewing mode or a low-pressure brewing mode.
6. The coffee brewing apparatus of claim 1, wherein: the brewing system is configured to alternately receive a plurality of cartridge holders which are each configured to receive one of a plurality of single serve capsules each having a different style or configuration.
7. The coffee brewing apparatus of claim 6, wherein: the plurality of cartridge holders is three cartridge holders; and the plurality of single serve capsules include a K-cup capsule, a Dolce Gusto capsule, and a Nespresso capsule.
8. The coffee brewing apparatus of claim 1, wherein: the grinding system includes a hopper for receiving the whole coffee beans, the grinding mechanism, and an adjustment mechanism for selecting a coarseness of the coffee grounds produced by the grinding mechanism.
9. The coffee brewing apparatus of claim 1, wherein: the grinding mechanism is one of a conical burr grinder, flat burr grinder, and blade grinder.
10. The coffee brewing apparatus of claim 1, wherein: the brewing system includes a brew head having an upper portion and a lower portion, the lower portion containing the brewing chamber, the upper portion being moveable relative to the lower portion between an open position and a closed position; and wherein the brewing system further includes a locking mechanism for maintaining a seal between a sealing element surrounding a water outlet in the brew head and the cartridge.
11. The coffee brewing apparatus of claim 1, further comprising: a drip tray slidably received by the housing and movable between a raised position and a lowered position.
12. The coffee brewing apparatus of claim 1, wherein: the coffee brewing apparatus is operable in both a high-pressure brewing mode to produce an espresso style coffee beverage, and a low-pressure brewing mode to produce a drip style coffee beverage.
13. A method for producing coffee, comprising the steps of: positioning a cartridge beneath a grinding mechanism of a coffee brewing apparatus; actuating the grinding mechanism to transform coffee beans into coffee grounds; collecting the coffee grounds in the cartridge; moving the cartridge with coffee grounds from a position beneath the grinding mechanism to a brewing chamber within a brew head located adjacent to the grinding mechanism; and initiating a brewing operation to cause heated water to pass through the coffee grounds within the cartridge.
14. The method according to claim 13, further comprising the step of: adjusting the grinding mechanism to select a coarseness of the coffee grounds produced by the grinding mechanism.
15. The method according to claim 14, further comprising the step of: adjusting the cartridge to one of a low-pressure brewing mode or a high-pressure brewing mode.

16. The method according to claim 13, further comprising the step of: placing a single-serve capsule containing coffee grounds in a capsule holder; positioning the capsule holder in the brewing chamber; and initiating a subsequent brewing operation to cause heated water to pass through the coffee grounds within the single-serve capsule.
17. The method according to claim 16, wherein: the single-serve capsule is one of a K-cup capsule, a Dolce Gusto capsule, and a Nespresso capsule.
18. A coffee brewing apparatus, comprising: a housing; a grinding system located on a front the housing, the grinding system having a grinding mechanism for grinding coffee beans to transform the coffee beans into coffee grounds; a cartridge holder located beneath the grinding mechanism for holding the cartridge beneath the grinding mechanism, for collection of the coffee grounds discharged by the grinding mechanism in the cartridge; and a brew head located on the front of the housing adjacent to the grinding system, the brew head including a brewing chamber for receiving the cartridge containing the coffee grounds, the brew head being movable between an open position allowing access to the brewing chamber for insertion and/or removal of the cartridge, and a closed position.
19. The coffee brewing apparatus of claim 18, wherein: the brewing chamber of the brew head is configured to alternately receive a plurality of cartridge holders which are each configured to receive one of a plurality of single serve capsules each having a different style or configuration.
20. The coffee brewing apparatus of claim 19, wherein: the plurality of cartridge holders is three cartridge holders; and the plurality of single serve capsules include a K-cup capsule, a Dolce Gusto capsule, and a Nespresso capsule.